Course Outcomes (COs)

- CO1: Explain the evolution of AI and deep learning and their role in Generative AI.
- CO2: Apply different Neural Network architectures for image, text and music generation.
- CO3: Design and optimize Generative AI models such as GANs and VAEs for real-world problems.
- CO4: Analyze the performance of different Generative AI models using BLEU, ROUGE and FID.
- CO5: Demonstrate awareness of ethical and legal implications of Generative AI.
- CO6: Demonstrate AI applications in different industry verticals.

Learning Objectives (LOs)

After successful completion of the Generative AI Laboratory, students will be able to:

LO1. Install, configure, and use modern Generative AI development environments, libraries, and frameworks such as **Python, TensorFlow, Hugging Face Transformers, and Google Colab** for AI application development.

LO2. Apply **Natural Language Processing (NLP)** techniques, prompt engineering strategies, transformer models, and Large Language Models (LLMs) to solve text generation and language understanding problems.

LO3. Design, implement, and evaluate **Generative AI models**, including **Variational Autoencoders (VAEs)** and **Generative Adversarial Networks (GANs)**, for image generation and reconstruction tasks.

LO4. Develop intelligent AI applications using **embeddings, vector databases, Retrieval-Augmented Generation (RAG)**, and pre-trained transformer models for semantic search and question-answering systems.

LO5. Analyze and evaluate the performance, reliability, and ethical aspects of Generative AI systems using appropriate evaluation metrics and responsible AI principles, including bias, hallucination, privacy, and fairness.

LO6. Design, deploy, and demonstrate end-to-end **Generative AI applications** using modern deployment frameworks such as **Gradio** or **Streamlit**, integrating the concepts learned throughout the laboratory.

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Exp. No.	Experiment Title	Aim	CO Mapping	PO	PSO
1	**Getting Started with Python, TensorFlow and Hugging Face Transformers**	To install and configure the Generative AI development environment and execute a pre-trained transformer model for text generation.	CO1	PO1, PO5, PO11	PSO1
2	**Text Preprocessing and Tokenization for NLP**	To perform text preprocessing, tokenization, and word embedding techniques for Natural Language Processing tasks.	CO1, CO2	PO1, PO2, PO5	PSO1
3	**Prompt Engineering using Large Language Models**	To design and evaluate effective prompts using Zero-shot, Few-shot, Chain-of-Thought, and Role Prompting techniques.	CO2	PO2, PO5, PO9	PSO1, PSO2
4	**Variational Autoencoders (VAEs) for Image Generation and Reconstruction**	To implement a Variational Autoencoder for image generation, reconstruction, and latent space visualization.	CO3	PO3, PO4, PO5	PSO1
5	**Data Visualization for AI Models**	To implement a Generative AI-based English-to-French text translation system using the T5 Transformer model, evaluate the generated translations using the BLEU score, and visualize the translation performance using different types of plots.	CO2,CO3	PO2, PO3, PO4, PO5, PO9	PSO1, PSO2
6	**Generative Adversarial Networks (GANs) for Image Synthesis**	To implement and evaluate a Generative Adversarial Network for synthetic image generation.	CO3	PO3, PO4, PO5	PSO1
7	**Transformer-based Text Generation using GPT and BERT**	To implement transformer-based language models and compare GPT and BERT for text generation and language understanding tasks.	CO2, CO3	PO2, PO4, PO5	PSO1
8	**Semantic Search using Embeddings and Vector Databases**	To perform semantic document retrieval using embeddings and vector databases such as FAISS or Chroma DB.	CO2, CO6	PO2, PO5	PSO1
9	**Retrieval-Augmented Generation (RAG) based Question Answering System**	To develop a Retrieval-Augmented Generation (RAG) application for document-based question answering.	CO3, CO6	PO3, PO4, PO5, PO8	PSO1, PSO2
10	**Responsible AI and Evaluation of Large Language Models**	To evaluate Large Language Models using BLEU, ROUGE, Perplexity, and analyze bias, hallucination, privacy, and ethical issues.	CO4, CO5	PO4, PO6, PO7	PSO2
11	**Deployment of a Generative AI Application using Gradio/Streamlit**	To develop and deploy an interactive Generative AI application using a pre-	CO3, CO4, CO6	PO2, PO3, PO5, PO8, PO9, PO11	PSO1, PSO2

trained Large Language Model and
Gradio or Streamlit interface.

CO PO and PSO mapping:

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	–	–	2	–	–	–	–	–	2	3	–
CO2	3	3	2	2	3	–	–	–	–	–	–	3	–
CO3	2	3	3	3	3	–	–	2	–	–	–	3	2
CO4	–	3	–	3	2	–	–	–	–	–	2	3	–
CO5	–	–	–	–	–	3	3	–	2	–	2	–	3
CO6	2	2	3	–	2	2	–	2	2	2	2	3	3

Experiment	COs Assessed
Exp. 1	CO1
Exp. 2	CO1, CO2
Exp. 3	CO2
Exp. 4	CO3
Exp. 5	CO3
Exp. 6	CO2, CO3
Exp. 7	CO2, CO6
Exp. 8	CO3, CO6
Exp. 9	CO4, CO5
Exp. 10	CO3, CO4, CO6